

Wenguan Zhang

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Target Position: LLM Application Developer



Education

Xidian University · School of Telecommunications Engineering 2024.09 – 2027.07

Master's · Information and Communication Engineering · ISN National Key Laboratory

North China University of Water Resources and Electric Power · School of Electronic Engineering 2020.09 – 2024.07

Bachelor's · Communication Engineering

Professional Skills

- Proficient in Java core fundamentals (collections, reflection, I/O, etc.) and concurrent programming (JMM, synchronized, volatile, CAS, thread pools, etc.); familiar with JVM runtime mechanisms (runtime data areas, class loading, and garbage collection).
- Familiar with MySQL basics including indexing, transactions, and locks; knowledgeable in database sharding and partitioning algorithms; familiar with Redis basic data types and persistence mechanisms.
- Familiar with mainstream Agent paradigms (Workflow, ReAct, Plan-Execute); knowledgeable in fine-tuning concepts and frameworks (LLaMA-Factory, PEFT).
- Knowledgeable in RAG-related concepts including vector embedding, query compression and rewriting, multi-path query expansion, semantic retrieval and metadata filtering, and text chunking strategies.

Project Experience

Intelligent Customer Service “Xiaofeng” 2026.02 – 2026.06

Project Overview: Based on a RAG architecture, implements a full pipeline from document ingestion, chunking, and vectorized storage to multi-source retrieval, intent routing, and streaming.

Tech Stack: Spring Boot + LangChain4j + Elasticsearch + Neo4j + Redis + MySQL + MinIO + MinerU

- **Document Parsing and Segmentation:** Introduced MinerU to support unified ingestion of multiple formats including PDF/Markdown/DOC/Excel; implemented a sliding window overlap + parent-child segmentation strategy to address semantic loss caused by fixed-size chunking, improving retrieval completeness from 75% to over 90%.
- **Intent Recognition Optimization:** Implemented 6-category intent classification and structured entity extraction based on LLM; improved intent recognition accuracy to 98% through CoT, Few-shot, and other optimization techniques.
- **Query Rewriting and Intelligent Routing:** Implemented 4-dimensional Query Rewriting (simplification/abstraction/error correction/normalization); query router dispatches queries to ES retrieval/MySQL/Neo4j with automatic fallback on failure.
- **Hybrid Retrieval and Re-ranking:** Built hybrid retrieval combining ES BM25 and vector semantic search, with RRF algorithm for multi-source ranking fusion; loaded BGE-RERANKER model via

ONNX Runtime for zero-network-latency re-ranking. Top-5 recall improved from 82% to 96%.

dodo-Agent

2025.11 – 2026.01

Project Overview: Adopts a multi-agent architecture to support web search Q&A (ReAct mode), file RAG Q&A (RAG + ReAct), intelligent PPT generation (state machine mode), and deep autonomous research (Plan-Execute-Critique mode).

Tech Stack: Spring AI + Project Reactor + MCP + Elasticsearch + MySQL + Redis + MinIO + Qwen

- **Deep Research Engine:** Implements a four-stage autonomous research paradigm, with each stage using dedicated role-based prompts (planning/execution/review expert) to ensure clear responsibilities; parallel/serial hybrid scheduling within plan-execute-critique loops based on task dependencies.
- **Dynamic Context Engineering:** Scans local file directories to read skill.md files and encapsulates skills as tools with progressive loading; designed an intelligent compression strategy that, when context exceeds limits across multiple iterations, retains user goals, tool execution results, and critique feedback by priority while discarding redundant conversation history.
- **Multi-instance Task Management:** Uses Redis distributed locks to ensure only one instance per session, Pub/Sub broadcasting to stop all instances, and TTL auto-renewal to prevent task state loss for long-running tasks.
- **Staged Streaming Output:** Implements SSE real-time push based on Project Reactor (Flux/Mono/Sinks), supporting four message types (thinking/text/reference/recommend) pushed to the frontend stage by stage.

Academic Achievements & Honors

- First Prize at the Provincial Level in the 15th Mathematics Competition
- Third-Class Freshman Scholarship for 2024 Master's Students
- Passed CET-4 and CET-6; strong English listening, speaking, reading, and writing skills